

MD KAMRAN CHOWDHURY SHISHER

Department of Electrical and Computer Engineering

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Education

Ph.D. in Electrical Engineering Auburn University, Auburn, AL Advisor: Prof. Yin Sun Thesis Title: Timely Inference over Networks	Aug. 2018 – May 2024
M.S. in Electrical Engineering M.S. in Electrical Engineering Auburn University, Auburn, AL	Aug. 2022
B.Sc. in Electrical and Electronic Engineering Bangladesh University of Engineering and Technology	Feb. 2013 – Sep. 2017

Research Interests

Communication Networks, Wireless Communications, Goal-oriented and Semantic Communication, Networked Control Systems, Machine Learning, Stochastic Optimization and Control

Professional Experience

Assistant Professor Department of Electrical and Computer Engineering University of Colorado Colorado Springs	Aug. 2026 – Present
Postdoctoral Researcher Department of Electrical and Computer Engineering, Purdue University Advisors: Prof. Christopher G. Brinton and Prof. Mung Chiang	May 2024 – Aug. 2026
Research Assistant Department of Electrical and Computer Engineering, Auburn University Advisor: Prof. Yin Sun	Aug. 2018 – May 2024
Teaching Assistant Auburn University	Jan. 2020 – May 2024
Lecturer Department of Electrical and Electronic Engineering Bangladesh Army University of Science and Technology	Nov. 2017 – June 2018

Awards and Honors (selected)

Auburn University Graduate School Distinguished Dissertation Award	2026
ECE Nominee for Purdue CoE Outstanding Postdoc Award	2026
The IEEE ComSoc William R. Bennett Prize	2025
Research Profile featured in ACM SIGMETRICS Performance Evaluation Review	2023
NSF Student Travel Grant, ACM MobiHoc	2023
NSF Travel Grant, North American School of Information Theory	2023

ACM SIGMOBILE Student Travel Grant, ACM MobiHoc	2022
IEEE INFOCOM Student Conference Grant	2022
NSF Student Conference Award, IEEE INFOCOM	2021
Dean's Award, BUET, Bangladesh	2017, 2016, 2015

Publications

Journal Papers: (Student Mentees Underlined)

- J7. **M. K. C. Shisher**, A. Piaseczny, Y. Sun, and C. Brinton, "Computation and Communication Co-scheduling for Multi-Task Remote Inference," under review in IEEE/ACM Transactions on Networking.
- J6. T. Z. Ornee, **M. K. C. Shisher**, C. Kam, and Y. Sun, "Remote Safety Monitoring: Status Updating for Situational Awareness Maximization," under review in IEEE Transactions on Information Theory.
- J5. C. Ari, **M. K. C. Shisher**, E. Uysal, and Y. Sun, "Goal-Oriented Status Updating for Real-time Remote Inference over Networks with Two-Way Delay," IEEE Transactions on Networking, vol. 34, pp. 4011-4025, 2026.
- J4. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang, "AoI-based Scheduling of Correlated Sources for Timely Inference," IEEE Transactions on Networking, vol. 34, pp. 2181-2195, 2026.
- J3. **M. K. C. Shisher**, Y. Sun, and I. Hou, "Timely Communications for Remote Inference," IEEE/ACM Transactions on Networking, vol. 32, no. 5, pp. 3824-3839, 2024. [[IEEE ComSoc William R. Bennett Prize \(2025\) Winner](#)]
- J2. **M. K. C. Shisher**, B. Ji, I. Hou, and Y. Sun, "Learning and Communications Co-Design for Remote Inference Systems: Feature Length Selection and Transmission Scheduling," IEEE Journal on Selected Areas in Information Theory, vol. 4, pp. 524-538, 2023.
- J1. K. Yan, **M. K. C. Shisher**, and Y. Sun, "A Transfer Learning-Based Deep Convolutional Neural Network for Detection of Fusarium Wilt in Banana Crops," AgriEngineering, vol. 5, no. 4, pp. 2381-2394, 2023.

Conference Papers: (Student Mentees Underlined)

- C12. Y. Zou, **M. K. C. Shisher**, V. Tripathi, and C. Brinton, "Communication-Efficient Personalized Adaptation via Federated-Local Model Merging," submitted, 2026.
- C11. Y. Zou, **M. K. C. Shisher**, V. Tripathi, and C. Brinton, "Distributed Online Convex Optimization with Interference Constraints: The Role of Gradient Freshness," submitted, 2026.
- C10. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang "Online Learning of Whittle Indices for Restless Bandits with Non-Stationary Transition Kernels," submitted, 2026.
- C9. A. Piaseczny, **M. K. C. Shisher**, S. Wang, and C. Brinton "RCCDA: Adaptive Model Updates in the Presence of Concept Drift under a Constrained Resource Budget," NeurIPS, 2025. [[Acceptance Rate: 24.52%](#)]
- C8. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang "AoI-based Scheduling of Correlated Sources for Timely Inference," IEEE ICC, 2025.
- C7. **M. K. C. Shisher**, A. Piaseczny, Y. Sun, and C. Brinton, "Computation and Communication Co-scheduling for Timely Multi-Task Inference at the Wireless Edge," IEEE INFOCOM, 2025. [[Acceptance Rate: 18.6%](#)]
- C6. C. Ari, **M. K. C. Shisher**, E. Uysal, and Y. Sun, "Goal-Oriented Communications for Remote Inference with Two-Way Delay," IEEE ISIT, 2024.
- C5. **M. K. C. Shisher** and Y. Sun, "On the Monotonicity of Information Aging," IEEE INFOCOM ASoI Workshop, 2024.
- C4. T. Z. Ornee, **M. K. C. Shisher**, C. Kam, and Y. Sun, "Context-aware Status Updating: Wireless Scheduling for Maximizing Situational Awareness in Safety-critical Systems," IEEE MILCOM, 2023.

- C3. **M. K. C. Shisher** and Y. Sun, “How Does Data Freshness Affect Real-time Supervised Learning?” ACM MobiHoc, 2022. [\[Acceptance Rate: 19.8%\]](#)
- C2. **M. K. C. Shisher**, H. Qin, L. Yang, F. Yan, and Y. Sun, “The Age of Correlated Features in Supervised Learning based Forecasting,” IEEE INFOCOM AoI Workshop, 2021.
- C1. **M. K. C. Shisher**, T. Z. Ornee, and M. F. Hossain, “QoS aware user association in massive MIMO enabled hetnets for DTU and NDTU traffic,” IEEE ICAEE, 2017.

Research Experience

- 1 **Postdoctoral Researcher** May 2024-Present
Purdue University
 - Optimized Machine learning (ML) model training methods for **resource-constrained edge devices** [C9] and designed a communication-efficient federated fine-tuning algorithm for **Large Language Models** on edge devices [C12].
 - Developed an **online resource allocation algorithm** for restless multi-armed bandits (RMABs) in unknown, non-stationary settings [C10].
 - Developed a **distributed convex optimization** algorithm under interference constraints [C11].
 - Developed **wireless scheduling algorithms for coupled or correlated sources** under interference constraints to minimize remote inference error [J6, C8].
 - Jointly optimized **communication scheduling and feature computation** to improve the performance of multi-task remote inference systems [J7, C7].
- 2 **Research Assistant** Aug. 2018-May 2024
Auburn University
 - Introduced a novel research topic on **Remote Inference**. Established an **information-theoretic theory** to interpret **the impact of information freshness on inference error** [C2, C3, C5] by using a generalized conditional entropy and by introducing the concept of ϵ -Markov chain and ϵ -**data processing inequality**.
 - Optimized the performance of remote inference systems by designing **scheduling algorithms** for both the single-user, single channel and the multi-user, multi-channel systems [J2, J3, C6, C3]. We proposed a new “**selection-from-buffer**” scheduling model to optimize the remote inference systems. Paper [J3] based on this research received [IEEE ComSoc William R. Bennett Prize \(2025\)](#).

Teaching Experience

1. **Guest Lecturer** Fall 2024
ECE 547: Introduction to Computer Communication Networks
Purdue University
2. **Teaching Assistant** Spring 2023, 2024
ELEC-5970/6970: Applied Statistical and Machine Learning
Auburn University
3. **Teaching Assistant** Spring 2020, 2021, 2022, Fall 2022
ELEC-7970: Reinforcement Learning
Auburn University
4. **Teaching Assistant** Fall 2022
ELEC-2120: Signals and Systems
Auburn University
5. **Lecturer** Nov 2017-June 2018
Bangladesh Army University of Science and Technology
 - Courses: Communication Theory, Digital Signal Processing, and Telecommunication Engineering

- Labs: Communication Engineering, Digital Signal Processing

Research Proposals

NSF VINES: Track 2: NextG Smart Factory/Manufacturing, submitted Sep. 2025

- **Contributions:** Thrust. Smart Factory/Manufacturing Operational Status.
- **PIs:** H. Kwon (Wichita State U), S. Kim (Iowa State U), M. Zhang (Mississippi State U); Senior Personnel include C. Brinton (Purdue U) and others from Ericsson Research, Penn State, Arizona State U, and UT Arlington.

NSF CPS: Medium: Name-Based Design of Distributed Intelligence for Smarter Communities, submitted June 2025

- **Contributions:** Thrust. Fine-Grained Control: Updating Models under Heterogeneity and Dynamics
- **PIs:** E. Yeh (Northeastern U), C. Brinton (Purdue U), S. Sannigrahi (Tennessee Tech), H. Newman (Caltech).

NSF PDaSP: Privacy-Preserving Aggregation of Demand Flexibility for Sustainable Power Systems, submitted Sep. 2024

- **Contributions:** Thrust. Tackling Hierarchical and Heterogeneous Environments
- **PIs:** J. Qin (Purdue U), A. Hashemi (Purdue U), C. Brinton (Purdue U), K. Poolla (UC Berkeley).

Student Mentoring

Ananth Ram Rajagopalan , Purdue University Graduate Student	Nov 2025-Present
Yinan Zou , Purdue University Graduate Student	Jan 2025-Present
Adam Piaseczny , Purdue University Graduate Student	May 2024-Present
Sam Chamoun , Auburn University Undergraduate Student	Aug. 2023-May 2024
Zachary Gayford , Auburn University Undergraduate Student	Jan 2024-May 2024
Cason B. Vazquez , Auburn University Undergraduate Student	Aug 2023-Dec 2023
Justin Tran , Auburn University Undergraduate Student	May 2021-May 2022
Cagri Ari , Middle East Technical University Graduate Student	Aug 2023-Present
Mengxue Li , Tuskegee University Graduate Student	Aug 2023-Present
Kevin Yan , Auburn High School Student	Aug 2022-Dec 2023

Talks

Computation and Communication Co-scheduling for Timely Multi-Task Inference at the Wireless Edge

- IEEE INFOCOM, London, UK May 2025

Timely Inference over Networks

- PhD Dissertation Defense April 2024
- Invited Talk at Purdue University, 2024 March 2024
- Invited Talk at Northwestern University, 2024 March 2024
- Invited Poster Presentation at ITA, 2024 Feb 2024
- Invited talk at Southeast Control Conference, 2024 Feb 2024
- Talk at Dept. of ECE, Auburn University Feb 2024

Learning and Communications Co-design For Remote Inference: Feature Length Selection and Transmission Scheduling

- Invited Talk at University of Maryland, College Park, MD Oct 2023
- Graduate Engineering Research Showcase, Auburn University Oct 2023

Communications of Timely Information for Real-time Machine Learning and Networked Intelligence

- North American School of Information Theory, Philadelphia, PA June 2023

How Does Data Freshness Affect Real-time Supervised Learning?

- Information Theory Application Workshop, San Diego, CA Feb 2023
- Auburn University Research Symposium (Poster Presentation) March 2023
- ACM MobiHoc, Seoul, South Korea. Oct 2022
- College of Engineering Research Showcase at the U.S. Space and Rocket Center, Huntsville, AL (Poster Presentation) Aug 2022

The Age of Correlated Features in Supervised Learning based Forecasting

- IEEE INFOCOM AOI Workshop, Vancouver, BC, Canada (virtual) May 2021

Services

Reviewer for Journal Manuscript Submissions

- IEEE Transactions on Information Forensics & Security, 2025
- IEEE Transactions on Signal Processing, 2025
- IEEE Transactions on Communications, 2024, 2025
- IEEE Transactions on Networking, 2024, 2025
- IEEE Transactions on Mobile Computing, 2024, 2025
- IEEE Transactions on Network Science and Engineering, 2024
- IEEE Transactions on Green Communications and Networking, 2024
- IEEE Journal of Communications and Networks, 2020, 2023
- IEEE Journal on Selected Areas in Information Theory, 2023
- IEEE Open Journal of the Communications Society, 2023
- IEEE Transactions on Wireless Communications, 2022, 2024
- IEEE Journal on Selected Areas in Communication, 2020

Reviewer for Conference Manuscript Submissions

- IEEE ITW, 2024
- IEEE WiOpt, 2024
- ACM MobiHoc ASoI Workshop, 2024
- IEEE ISIT, 2022
- IEEE INFOCOM, 2020, 2022
- IEEE WCNC, 2021, 2022
- IEEE INFOCOM, 2020
- IEEE INFOCOM AoI Workshop, 2019, 2020

TPC member

- IEEE WCNC, 2026
- IEEE WCNC, 2022
- IEEE WCNC, 2021

Conference Session Chair

- IEEE INFOCOM ASoI Workshop, 2025
- IEEE INFOCOM DTWin Workshop, 2025

Maintainer of an online paper repository on Age of Information

Aug 2018-May 2024

Volunteer on E-Day, Auburn University, Auburn, AL

Feb 2020

Organizing Secretary, Bangladesh Student Organization, Auburn University

Aug 2022-Aug 2023

Professional Membership

IEEE Member

IEEE Information Theory Society Member

IEEE Communication Society Member

ACM SIGMOBILE Member